

Examiner reports

Q1.

This question gave students complete choice as to how to go about solving the problem. Many good attempts were seen, with the full range of marks being scored. Different correct approaches were used and given credit. For example, some students modelled using the length and width of the rectangle, others used a radius and angle to form an expression for area using trigonometry.

Marks were often lost through students not fully justifying their answer. Some students failed to identify their variables or were inconsistent in their use. Others did not explain why they had differentiated or how they knew they had found a maximum.

Q2.

This question saw most students making some progress, with a full range of marks being awarded. A high proportion of students realised that implicit differentiation should be used, but very few differentiated efficiently. Instead, the approach seen most frequently was to first expand the brackets and then differentiate. This resulted in more work to find the required solution. The most frequent mark seen for this question was 7/7. Students who made mistakes at the beginning were still rewarded for using correct techniques.

Q3.

In (a), most students realised that the domain would take values $x > 1$ in this question. However, in order to gain full marks it was necessary to state the domain in a complete form involving the notation $x \in \mathbb{R}$, which most students failed to do.

In (b), a large number of students started to differentiate the printed result for $f'(x)$, rather than differentiate $f(x)$. Many then realised their mistake and restarted the question on a continuation sheet, with a good number scoring full marks.

There was a wide spread of marks across question (c). Most students understood that they needed $f''(x) = 0$, though a number instead used $f'(x) = 0$. Of those students who successfully solved the correct equation, very few scored full marks, as they either failed to correctly deal with the solution $x = 1$ or failed to test either side of their correct solution $x = 4$. Only a small proportion of students identified the correct interval in (d).

Q8.

In part (a), apart from a small number of students who treated the expression as a product of terms instead of a sum and hence gained no marks, this was usually answered correctly. Where there were errors, it was usually with the first term as the students wrote e^{3x} or $3e^x$ instead of $3e^{3x}$.

In part (b)(i), most students applied the quotient rule correctly. However, poor notation such as $1 + \cos^2 x$ instead of $(1 + \cos x)^2$ or $\sin X^2$ instead of $\sin^2 x$ and sign errors caused marks to be lost. There were also examples of false cancellation leading to the correct result. In part (b)(ii), here the students tended to do either very well or very badly, scoring 2 or 0. Those students attempting to use $y = \ln(\sin x) - \ln(1 + \cos x)$ rarely made progress.

Q9.

In part (a), apart from small number of students who did not differentiate the expression as

a product of two functions and obtained gradient functions such as $\frac{dy}{dx} = -8x \sin 2x$, hence gaining no marks, this question was usually answered very well. Some students

failed to simplify their expression for the gradient after substitution of $\frac{\pi}{4}$, thus giving rise to some very complicated forms for the equation of the line.

In part (b), almost all the students demonstrated they could integrate by parts, with many gaining full marks. Where marks were lost it was usually due to an incorrect coefficient

written in the integration of $\frac{dy}{dx} = \cos 2x$ with $v = 2\sin 2x$ being common. Some students failed to simplify their final expression and there were some cases where a non-exact decimal value was written as an answer. A small number of attempts at the volume instead of the area were seen.

Q10.

This question was well answered by the great majority of the students.

- (a) Any errors that occurred were usually associated with the second term where $\tan 2x$ was differentiated to $\sec^2 2x$ or $2\sec^2 x$ instead of $2\sec^2 2x$. Attempts at using the

quotient rule on $y = \frac{x^4 \sin 2x}{\cos 2x}$ was often prone to errors.

- (b) Most students used the quotient rule correctly, although some had the two terms in the numerator reversed. Those who chose to use the product rule were mostly correct, with a few making a sign error. A small number of students failed to go on

substituting $x = 3$ into their derived equation. A common incorrect answer was $\frac{3}{2}$; arising from students expanding the denominator incorrectly to $x^2 - 2x - 1$.

Q11.

In part (a), it was essential to use the quotient rule, as requested – it is specifically mentioned in the specification – and to show sufficient steps in the proof of the given answer. It was surprising that quite a number of candidates were unable to go from

$\frac{-\cos \theta}{\sin^2 \theta}$ to the final expression.

In part (b), it was good to see some completely correct solutions though some used 30° and 45° , which was incorrect. A substantial number of candidates who correctly started the question lost a minus sign in their work.

Many candidates fell at the first hurdle by not replacing dx by an expression in θ and $d\theta$.

There were B marks available for a correct $\frac{dx}{d\theta}$, for $\sqrt{\operatorname{cosec}^2 \theta - 1}$ converted to $\sqrt{\cot^2 \theta}$ and for the limits in radians, but even these marks were often not earned. Unfortunately,

many confused $\frac{dx}{d\theta}$ with $\frac{d\theta}{dx}$ and of course made no headway after that. Candidates who

produced the correct numerical answer from their calculators without correct working did not gain marks.

Q12.

This question was in general a good source of marks for the majority of students. The main cause of loss of marks was a common desire to use an approximate numerical value for e .

Q13.

Part (a) proved to be a good source of marks, although a significant number of students lost a mark through poor notation, such as $\cos y^2$ or no $\frac{dx}{dy}$

Although many earned full marks in part (b), many other proofs were inadequate. Students were expected to use the identity $\tan^2 y + 1 = \sec^2 y$ and to show the expansion of $(x - 1)^2$ clearly. Too many fudged their working, wrote $\tan y + 1 = \sec y$ or only proved the known trig identity.

The result in part (c) was easily derived from the previous two parts, but there were many blanks here and many confused statements about inverse functions which were often

taken as $\frac{1}{\tan y}$. Students who used the formula book appropriately generally earned the mark.

There were many fully correct solutions to part (d)(i). However it was distressing at this stage to see some horrendous algebra, with terms inverted in ways that were invalid and worthless.

Only the more able students were able to make progress on the last two parts of the question. Many students differentiated their quadratic instead of the expression for $\frac{dy}{dx}$. Some students earned the method mark in the final part, but few completed the whole answer, as their final explanation often had an element missing.

Q14.

Part (a) was well answered by the majority of candidates with many fully correct responses seen. Most candidates used substitution of $u = (x^3 - 1)$ to give $y = u^6$. Although $(x^3 - 1)^5$ was usually seen, errors did occur with the associated function of x .

Part (b)(i) was very well answered, with a majority of candidates earning both marks. The major error was from candidates who did not treat this as a product and simply differentiated x to get 1

and $\ln x$ to get $\frac{1}{x}$.

Part (b)(ii) Again, this part was very well answered. Many fully correct responses were seen. The main error was with candidates who wrote the gradient as $1 + \ln e$ which earned the method mark. Then this was often evaluated as $1 + \ln e = 1$ or left unsimplified. Several candidates who

obtained $\frac{dy}{dx} = 2$ then went on to use a gradient of $-\frac{1}{2}$.

Q15.

In part (a)(i) it was clear that dx/dy confused many, they simply wrote down $dy/dx =$. Many solutions were correct, but some candidates missed the multiplier and some were clearly trying to use the product rule or split $3y$ and $+1$, which was an indication of poor understanding of work on trigonometric functions.

Part (a)(ii) was not very well answered by many candidates. Those who had managed to get the

method mark in part (a)(i) often gained the method mark for obtaining $\frac{dy}{dx} = 3\sec^2(0)$. However,

many incorrectly rearranged the function to $\frac{dy}{dx} = \frac{1}{3\cos^2(3y+1)}$ not $\frac{\cos^2(3y+1)}{3}$, so substituting into an incorrect function

In part (b) a large number of candidates had no idea of the shape of the inverse tangent graph. A few incorporated a turning point and many failed to put the correct values on the y -axis. For the correct shapes that were seen, the y -axis was often not labelled, so only part marks were awarded. The main error seen was a reflection of $y = \tan x$ in the y -axis and not in $y = x$.

Q16.

In part (a)(i) the majority of candidates were able to differentiate e^{2x} and apply the product rule correctly but a few made an error in one or other of these processes. Occasionally an attempt to simplify was incorrect.

In (a)(ii) when finding the equation of the tangent (a line) it is essential to find the gradient (a constant) at the requisite point first. Most did this but it is disconcerting, at this level, to find candidates who give a non-linear equation as their answer by failing to do the first step. A few added $e^2 + 2e^2$ wrongly or lost exactness by evaluating $3e^2$, and a few found the equation of the normal.

In (b) although the quotient rule was known by almost everyone, and almost all earned the first mark, some appalling algebra and trigonometry abounded thereafter. A few failed even to get the first mark as they put a term in the wrong place on the numerator, had $\sin x$ or $\cos x$ instead of $\sin 3x$ or $\cos 3x$ in one of their differentials or else their denominator was $1 + \cos^2 3x$ instead of

$(1 + \cos 3x)^2$. A few failed to use the chain rule when differentiating $\sin 3x$ or $\cos 3x$. Those who missed the brackets around $(1 + \cos 3x)$ in the numerator sometimes recovered but far too many falsely cancelled $\cos 3x$ at this stage. If the numerator was correctly expanded another mark was available, although $\cos 3x^2$ and $\cos^3 3x$ instead of $\cos^2 3x$ for $(\cos 3x)^2$ was common; this was recoverable, but $\cos 9x^2$ was not. After that more incorrect cancellation was seen. Taking out a common factor of 6 was the safest route forward, as those who tried to deal with $6\cos^2 3x + 6\sin^2 3x$ often put it equal to 1 or 18 instead of 6. Even being correct to that stage did not ensure completion as, again there was false

cancellation. Those who split the terms, then used $(1 - \cos^2 3x)$ for $\sin^2 3x$ and factorised to $(1 - \cos 3x)(1 + \cos 3x)$ generally completed successfully. It was quite extraordinary how many methods with completely wrong working

landed up at $\frac{6}{1 + \cos 3x}$.

Q17.

In part (a) about a third of the candidates could not evaluate the expression in e correctly, the

term $e^2 \left(\frac{1}{e}\right)^2$ causing many problems.

In part (b) most candidates scored the marks for the implicit differentiation, with many using the product rule correctly. The common errors were to keep the y in the derivative, of $2y$, or not to have two terms in the product or to drop a 2. Although virtually all candidates differentiated x^2 correctly, the constant in terms of e caused problems for many; they “differentiated” it despite being told it was constant in the question. Most attempted to solve their expression for

$\frac{dy}{dx}$ and of

the many who had incorrectly written $\frac{dy}{dx} =$ in their opening line of implicit differentiation, most ignored it to get a correct expression, and thus it was condoned. Many were unable to obtain a fully correct solution as they still had their constant term, C in terms of e , in their expression.

In part (c) candidates had to have the expression for $\frac{dy}{dx}$ correct from part (b) in order to achieve both marks. Most substituted the given values of x and y into their expression and knew it should come to zero, and stated it was zero even if it clearly wasn't, rather than looking for an error. About 30% of the candidates gave a fully correct solution to the whole question. A few who otherwise had correct solutions, failed to bring the solution to a conclusion, although most did state a stationary point required the derivative to be zero, or equivalent.

Q18.

Part (a) was well answered by the majority of candidates. Many fully correct responses were seen and, if there were errors, it was usually the final accuracy mark that was lost through the omission of brackets at some stage in the solution.

Part (b) was not answered as well as part (a). Although many candidates were successful in factorising the required quadratic, there were also many solutions which were accompanied by terms in e^{-4x} . Where marks were lost, it was mainly due to incorrect signs, though some candidates did manage to obtain follow through marks.

A few candidates attempted to substitute their values of x back into the derivative rather than y . Many candidates stopped when they had found the two values of x , presumably

thinking they had finished the question.

Q19.

Part (a) was well answered with many candidates gaining full marks. Where part marks were earned, the most common error was to lose the factor of 4 in the numerator by incorrectly writing the derivatives of $\sin 4x$ and $\cos 4x$ as $\cos 4x$ and $-\sin 4x$.

Many candidates did not attempt part (b). Although fully correct responses were seen, this was the question part where most candidates scored very few if any marks. Where candidates split the term $4\tan^2 4x$ up into $(4\tan 4x)(\tan 4x)$ or similar expressions they seemed to have more success. In general, the use of the chain rule from whatever starting point was not coped with satisfactorily.

Q20.

There were many correct answers to this question, although the amount of working shown varied considerably; some, for example, went into u and v for the product whilst others just wrote all the derivatives down. Those who had the derivatives all correct usually went on to calculate the gradient correctly, although mistakes were seen in handling $e^0 - 1$.

The common error in the differentiation was in the e term where the $\frac{dy}{dx}$ was not attached or it became ye^0 or similar. Some candidates differentiated the expression with respect to y , and inverted the result for the gradient, which was both perceptive and, of course, acceptable. Some candidates thought they should take logs of both sides first, and often went on to create nonsense, although those who had recognisable derivatives in an attempt at the chain rule could gain partial credit.

Q21.

In part (a)(i), most candidates stated the formula for $\sin 2\theta$ correctly although not always in its simplest form.

In part (a)(ii), although many candidates knew the formula for $\cos 2\theta$ in at least one its correct forms, several gave $\cos^2 \theta + \sin^2 \theta$, and later in the question just gave the value of $\cos 2\theta$ as 1 without apparently thinking something could be wrong.

In part (a)(ii), many candidates had problems. It appeared that the 3-4-5 triangle was either not well known or not recognised as relevant; although some candidates did draw this to justify their answer others used $\sin^2 \theta = 1 - \cos^2 \theta$. Many candidates worked backwards from the answer given, to find a value for $\sin \theta$, which gained no credit nor did going via an angle from the given value of $\cos \theta$. Many candidates omitted to find $\cos 2\theta$ or, as above, gave the value as 1, or another incorrect value from an incorrect expression. Those who did find the correct value for $\cos 2\theta$ often dropped the minus sign in later working.

In part (b)(i), the use of the chain rule with these parametric equations was generally done well with most candidates completing it correctly. Some candidates apparently tried to integrate rather than differentiate and some made sign errors when differentiating. A few candidates had the chain rule expression the wrong way up.

In part (b)(ii), although some neat and concise correct answers were seen, many candidates got in a mess trying to use angles and not working in exact fractions. Some confused the value of the sine and cosines involved with the angles writing terms such as $\sin \frac{3}{5}$. Some left a mix of fractions and decimal numbers in their final answer, which is bad practice but was accepted if the values of the coordinates at P were correct.

Q22.

- (a) (i) The first part of this question was reasonably answered with many candidates obtaining both marks. Where candidates obtained 1 mark it was because many ended up with the answer $1/(5x - 2)$.
- (ii) Again this part was very well answered with candidates obtaining both marks. The majority of candidates arrived at $K \cos 2x$ but $K = -2$ was a common error.
- (b) (i) This part was not answered very well. Many candidates lost a mark through using -0.693 instead of $\ln \left(\frac{1}{2} \right)$ and $f(x) \geq 0$ was a common response.
- (ii) Most candidates were able to do this part with the correct answers often seen. The main error was the omission of brackets around $5x - 2$ obtaining $\sin 2(\ln 5x - 2)$. The expression for $fg(x)$ was also often seen.
- (iii) For those candidates with a correct starting expression many went on to get full marks. Those candidates who used $\sin \ln (5x - 2)^2$ often lost an accuracy mark for not rejecting one of their answers. Most candidates obtained the first method mark for making the correct initial step for their expression.
- (iv) This was usually well done but there were common errors of dividing by $\sin 2$ obtaining $y/\sin 2 = x$ or even $y/\sin = 2x$ and $y/2 = \sin x$

Q23.

Many candidates lost marks on this question from careless work when handling the number of e^{2x} terms involved.

- (a) Most candidates obtained the correct coordinates although $(\ln x, 0)$ was a common error.
- (b) This was generally well answered by those candidates able to use the quotient rule. Some errors with the numerator being reversed to $(\ln x - 1)$ meant candidates lost accuracy marks later in the question. Many candidates obtained 3 marks for reaching $\ln x - 1 = 0$ and many went on to find $x = e$. However many stopped here without finding the value of y .
- (c) Although many correct solutions were seen some candidates lost marks through leaving $(e^{3x})^2$ rather than simplifying it. There was also a large number of candidates who did not answer part (b) but started again here, used the quotient rule correctly and went on to then answer the remainder of the question correctly. They seemed to realise that they needed the derivative to find a gradient but not for finding stationary points

Q24.

In part (a), most candidates used the product rule correctly, but at this stage $e^{2x} \neq 0$ was rarely seen, although many candidates obtained the correct quadratic equation. Factorisation, or use of the formula, was usually done well and many candidates obtained both of the correct solutions. Where candidates seemed unsure of what to do with the e^{2x} , many expressions involving the use of $\ln e^{2x} = 2x$ were seen together with the formation of a cubic expression in x . A third value of $x = 0$ was also seen by candidates who thought this was the solution of $e^{2x} = 0$.

Part (b) was generally well answered by those candidates able to do the product rule in part (a). Some errors with the coefficients meant candidates lost accuracy marks.

In part (c), although many correct solutions were seen some candidates lost marks

through incorrect evaluation of $\frac{d^2 y}{dx^2}$.

Q25.

Candidates were generally successful in part (a), sometimes aided by the allowances made for $(-1, 180)$.

Part (b) was well answered by many candidates. Common errors were the use of 'shift' and 'transformation' to describe the word 'translation', and the stretch was often given a scale factor of $\frac{1}{2}$ in the x -axis.

In part (c), the shape of the graph was usually correct, but many candidates lost this mark by going into other quadrants. The mark for the labelling of the axis was not so readily earned, as these values were often missing or incorrect.

In part (d)(i), many candidates gave the correct response. The major error was to state that $2y = \cos^{-1}(x - 1)$ for the first step resulting in no marks being awarded.

In part (d)(ii), most candidates at least obtained the method mark. Where candidates correctly differentiated, they went on to achieve full marks.

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Q26.

Very few candidates achieved completely correct solutions to this question.

Part (a) was well answered by the majority of candidates. The main error was the two terms in the numerator being interchanged, resulting in $k = +12$. Other errors came from the expansion of the brackets, with various numerators including $8x$, and with -3 not being multiplied by 4, giving a final constant of -3 .

In part (b)(i), the expression $\ln(4x - 3)$ was rarely differentiated completely correctly. A

common error was to get $\frac{x}{4x-3} + \ln(4x-3)$, rather than $\frac{4x}{4x-3} + \ln(4x-3)$, which earned two method marks.

In part (b)(ii), $\frac{dy}{dx}$ was frequently calculated correctly, following through from part (b)(i),

but then this value was used as the gradient of the normal, with some candidates explicitly stating that they were finding the normal.

In part (c)(i), most of the candidates earned a method mark for ' $du = 4dx$ '. Many of these candidates went on to write the integral completely in terms of u . Some candidates lost

marks at this stage through either obtaining an extra factor of 4 or getting $x = \frac{u-3}{4}$ when rearranging $u = 4x - 3$.

Although many fully correct responses were seen, some candidates were unable to proceed beyond this point: part (c)(ii) proved to be beyond many candidates, as the correct application of integration by parts was not common. Most candidates started the

question with $u = \ln$ and $\frac{dv}{dx} = 4x - 3$. Those candidates who correctly started this part of the question often failed to get the first accuracy mark because of their derivative of $\ln(4x - 3)$, losing the factor of 4 in the numerator. However, many were able to get the next method mark because they correctly used their value from part (c)(i). A few candidates spotted that the result obtained in part (b)(i) could provide an alternative method of solution.

Q27.

Part (a)(i) was not very well answered by the majority of candidates. Fully correct responses were seen but it was usually from candidates who successfully rearranged the equation into the form $f(x) = 0$, which was seen in a number of acceptable forms. Where candidates simply substituted in the two values given into the LHS they obtained 1 and 0 but still indicated that this was a change of sign and therefore there was a root. Most candidates achieved at least one mark but lost the second by simply stating change of sign therefore a root.

Very few incorrect responses were seen in part (a)(ii).

Part (a)(iii) was very well answered. The main error was with the candidates who used degrees rather than radians

Part (b)(i) was very well answered, with most candidates successfully using the quotient rule. Where errors occurred it was usually through missing brackets, although many candidates were able to recover at some stage in the working.

As part (b)(ii) followed on from part (i) most candidates were able to obtain the mark for substitution; even those who had incorrectly simplified their work. Many fully correct responses were seen. Several candidates also stopped after the substitution of $x = 0$ and left their answer as -2 .

Q28.

It was pleasing to see complete correct solutions from a large number of candidates. Clearly implicit differentiation of equations such as this is a topic that had been learnt well. It was a pity that many candidates wasted time in rearranging the equation in x , y and dy/dx to isolate dy/dx . A quicker method was to substitute $x = 1$ and $y = 3$ immediately after differentiating. A few candidates simply wrote $8xdx + 2ydy = 3ydx +$

$3xdy$ or similar, which is incorrect notation and scored no marks.

The common errors were to leave the 4 in the differentiated expression, or not to differentiate the product term correctly, attaching dy/dx to the wrong term or having just one term. The few candidates who chose to rearrange the given equation and solve it for y before differentiating generally made little further progress, although such methods would be valid if carried out correctly.

Q29.

Part (a)(i) was well answered by the majority of candidates. Many fully correct responses were seen, and, if there were errors, it was usually through further incorrect work or by the omission of brackets.

Very few incorrect responses were seen to part (a)(ii). Most candidates appeared to be able to use the product rule successfully. Some errors with signs were made, leading to the loss of the accuracy mark.

Part (b) was very well answered.

Q30.

Many candidates showed that they are competent with implicit differentiation and scored full marks here or at least some of the marks. The most common error was for candidates not to differentiate the 4 on the right-hand side, leaving it as 4, having got the left-hand side correct. The derivative of $2y^2$ was usually correct, but errors such as an extra x or y or a coefficient error crept into the derivative of $3xy$.

Q31.

Part (a) was well answered by the majority of candidates. Many fully correct responses were seen and if there were errors it was usually for failing to multiply by the derivative of $(3x + 1)$ so answers of $5(3x + 1)^4$ were not uncommon.

Part (b) was not as well answered as part (a) but many correct responses were seen. The main error was again to miss the factor of 3 and the very common incorrect answer was $1/(3x + 1)$.

In part (c) most candidates used the product rule successfully but a number of candidates believed that the differential of a product was the product of the differentials.

Candidates who made errors in part (b) were able to recover and earn 2 marks. A very common incorrect response was to simply multiply the answers to parts (a) and (b) resulting in $45(3x + 1)^3$. Another common error was to 'simplify' $\ln(3x + 1)(3x + 1)^4$ to obtain $\ln(3x + 1)^5$.

Q32.

Part (a) of this question was reasonably answered with most candidates obtaining the method mark for the product rule. Some candidates then lost the accuracy mark through incorrect evaluation of the constant associated with the derivative of $\cos 2x$; 2 and $\frac{1}{2}$ were frequently seen.

Answers to part (b)(i), although frequently correct, were often badly set out with the function being equated to zero or x being changed to α at various points in the solution. Division of $\cos 2x$ by $\cos 2x$ often resulted in zero before going on to fudge the correct answer. Part (b)(ii) was usually well answered with correct evaluations of $f(0.4)$ and $f(0.5)$. Some candidates then lost marks by just saying α was 'between these two values' without stipulating 0.4 and 0.5. Part (b)(iii) was usually well done but there were many cases when

the tan became removed from its $2x$ and $\frac{1}{2} x \tan^{-1} = x$ was often seen. Part (b)(iv) was generally well answered but marks were lost from the use of degrees and for not writing the answer to the required degree of accuracy.

Many fully correct answers and many answers which only lost the final accuracy mark were seen in part (c). Other answers which had the wrong coefficient associated with the $\sin 2x$ often got the method marks. There were candidates who started with $u = x$ and $v =$

$\cos 2x$ or started with $\frac{dv}{dx} = x$ and obtained a more difficult integral.

Q33.

Part (a) was very well answered with most candidates gaining all 3 marks but it was not uncommon to see an incorrect quotient formula used (usually the wrong order in the numerator, but sometimes the wrong denominator)

Part (b) was reasonably well answered. However a number of candidates tried to tackle this question by squaring, generally without success.

There were very few successful attempts seen in part (c). Most candidates failed to provide $\frac{dx}{d\theta}$ at all. Many of those who did make an attempt chose to try a different substitution of their own, either immediately or subsequently.

Q34.

In part (a)(i), many candidates were unable to produce a correctly expressed solution, despite dealing correctly with the differentiation. The main error was the omission of brackets around

$8x + 3$. Part (a)(ii) was well answered by the majority of candidates. Part (b)(i) was well

answered by the majority of candidates, although a common error was $6y^2 + \frac{1}{x}$. Most candidates correctly substituted $y = 1$ and obtained 7, although 6 was a common evaluation; a substitution of $y = 2$ was also seen. Many candidates who obtained 7 then proceeded, incorrectly, to write $(y - 1) = 7(x - 2)$ although a few candidates correctly wrote

" $(x - 2) = 7(y - 1)$ ". The significance of $\frac{dx}{dy}$ was rarely appreciated, with a significant

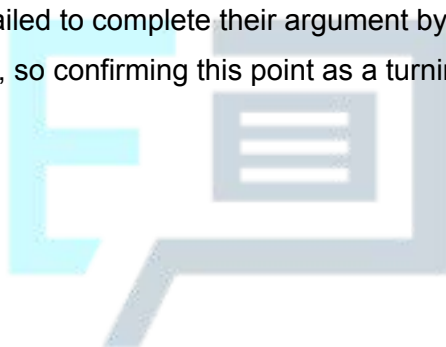
number of candidates finding an incorrect gradient of $-\frac{1}{7}$, and confusing $\frac{dx}{dy}$ with the gradient of a normal.

Q35.

The responses to this question were very mixed, varying from the very poor to some excellent answers. Most, but not all, candidates did attempt to separate and integrate in part (a)(i) with some making an error in their integrals, and rather more omitting a constant. Such candidates could make no further progress in part (a)(ii) without a constant to find. Some candidates thought to bring e^{-1} into their answer in part (a)(i), instead of the expected conventional constant, and got themselves confused in part (a)(ii) as they could not convincingly obtain the given answer. Some candidates made the anticipated error at the exponentiation stage of $\ln y = -\cos t + c \Rightarrow y = e^{-\cos t} + e^c$. Many candidates could not justify the 50, other than saying it was the initial value.

For part (b) many candidates realised they had been given the required expression in part (a) and substituted and evaluated correctly, although some ignored the request to answer to the nearest centimetre, any greater accuracy being rather meaningless in this modelling problem. Some used degrees instead of radians. Other candidates substituted into the differential equation. In part (b)(ii), although most candidates knew in principle what to do, they could not find the second derivative correctly. Many treated y as a constant or as if it were t , rather than using implicit differentiation.

Some candidates did attempt to differentiate the given answer in part (a)(ii) twice, but usually made an error in their use of the chain rule. Some candidates did get the differentiation correct, but failed to complete their argument by demonstrating that the first derivative was zero at $t = \pi$, so confirming this point as a turning point.



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